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Translational friction coefficients of rigid, symmetric top macromolecules. Application to circular cylinders

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The hydrodynamic interaction equations and the interaction tensor that appear in the theoretical formalism for calculating translational friction coefficients of rigid particles modeled as assemblies of beads, are formulated in cylindrical coordinates. For symmetric top particles, the cylindrical shielding tensors of symmetry-equivalent beads are the same, and have only five nonzero components. This effects a great simplification with respect to the Cartesian formalism. Application is made to the end-effect correction, for translational motion of right circular cylinders modeled as stacks of rings, extrapolated to zero bead size. No appreciable changes in the translational friction coefficients are found when the hollow cylindrical models are capped at the ends with disc-shaped plates. For finite cylinders with axial ratio $p > 2$, our results, as well as those from other theoretical and experimental studies, are in remarkable disagreement with the previous calculations by Broersma.

INTRODUCTION

During the past few years, a general method for calculating hydrodynamic properties of rigid macromolecules has been developed by several workers.¹⁻⁸ In this method, the particle is modeled as an array of N spherical elements and the hydrodynamic interaction between them is obtained by solving a system of linear equations, whose matrix of coefficients is of dimension $3N \times 3N$. When N is high enough (say $N > 100$), the solution of such a system becomes practically impossible because of computer time and memory limitations. However, a good reproduction of the details in macromolecular shape is only achieved when a large number of small elements is used. The shell-type modeling proposed by Bloomfield *et al.*⁹ is the ideal limiting case, and has been shown to give best results.^{9,10}

Rigid macromolecules and macromolecular complexes often exhibit some degree of symmetry that could lead to great simplifications if it were properly taken into account. In many cases, the particle has an axis of symmetry plus a symmetry plane containing this axis. This is the case for all rodlike (cylindrical) particles, multisubunit proteins such as hemocyanin, and macromolecular complexes such as microtubules and T -even bacteriophage.

It is the purpose of this paper to introduce the elements of symmetry mentioned above in the calculation of a simple hydrodynamic property, the translational friction tensor. To do so, the modified interaction tensor^{11,12,3} and the set of equations describing the hydrodynamic interaction are expressed in cylindrical coordinates referred to the symmetry axis. Thus, the problem is reduced to evaluating the contributions from sets composed of symmetry-equivalent elements, rather than those from each individual element.

As an important application of the new method, we present a study of the translational friction tensor of right circular cylinders. Up to now, the standard calculation has been that by Broersma,^{13,14} who used an extended version of Burgers' method¹⁵ to obtain the end-effect correction. Recently, Swanson *et al.*¹⁰ have obtained the translational friction coefficients of short

cylinders with length-to-diameter ratio $p = 1, 2$, and 4 by extrapolation to zero bead size of results for cylindrical models composed of spherical beads. Since they worked in Cartesian coordinates, their calculation was restricted to small values of p , owing to computational limitations. Nevertheless, a significant discrepancy of their results with respect to Broersma's was found.

We have carried out calculations of the end-effect corrections for both lengthwise and sidewise motion of moderately long cylinders. The results have been extrapolated to the shell-model limit and, tentatively, to $p \rightarrow \infty$. As will be shown later, symmetry considerations allow us a dramatic increase in the number of elements composing the cylindrical models, so that a wide range of p can be studied, and the extrapolations are supposed to be reliable. We have found that the disagreement with Broersma's end-effect corrections persist when p is as high as, say, 100, although his values for $p \rightarrow \infty$ might be correct.

THEORY

General formulation

The hydrodynamic interaction equations for a system composed of N spherical elements are given by

$$\mathbf{v}'_i + \sum_j \mathbf{T}_{ij} \mathbf{F}_j = \mathbf{v}_i, \quad i = 1, 2, \dots, N, \quad (1)$$

where $\mathbf{v}_i$ is the unperturbed relative velocity and $\mathbf{v}'_i$ is the actual relative velocity, including the hydrodynamic interaction effect. $\mathbf{T}_{ij}$ is the modified hydrodynamic interaction tensor for unequal elements,³

$$T_{ij}^{\alpha\beta} = \frac{1}{8\pi\eta_0 R_{ij}} \left[c_1 \delta_{\alpha\beta} + c_2 \frac{R_{ij}^\alpha R_{ij}^\beta}{R_{ij}^2} \right], \quad (2)$$

$$\alpha, \beta = x, y, z,$$

where $\delta_{\alpha\beta}$ is Kronecker's delta and

$$c_1 = 1 + (\sigma_i^2 + \sigma_j^2)/3R_{ij}^2, \quad (3)$$

$$c_2 = 1 - (\sigma_i^2 + \sigma_j^2)/R_{ij}^2. \quad (4)$$

σ_i the radius of the i th bead, η_0 the solvent viscosity, and R_{ij} the vector joining elements i and j . Now we in-

roduce cylindrical coordinates (θ_i, r_i, z_i) for the beads' positions. To distinguish between the Cartesian and cylindrical representations, the superscript c will be hereafter attached to vectorial and tensorial quantities when they are expressed in cylindrical coordinates; otherwise, Cartesian coordinates are assumed. Thus,

$$\mathbf{v}_i = (v_i^x, v_i^y, v_i^z)^T \quad (5)$$

and

$$\mathbf{v}_i^c = (r_i \omega_i, \rho_i, v_i^z)^T. \quad (6)$$

ω_i and ρ_i in Eq. (6) are, respectively, the angular and radial velocities. The superscript T denotes the transpose. If $\mathbf{A}_i$ is the coordinate transformation tensor for the i th bead

$$\mathbf{A}_i = \begin{pmatrix} -\sin\theta_i & \cos\theta_i & 0 \\ \cos\theta_i & \sin\theta_i & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (7)$$

then

$$\mathbf{v}_i = \mathbf{A}_i \mathbf{v}_i^c. \quad (8)$$

The above equations also hold for the perturbed (primed) velocities. Since the frictional elements are spheres, their frictional coefficients being given by $\zeta_i = 6\pi\eta_0\sigma_i$, we can formulate similar equations for forces. Thus,

$$\mathbf{F}_i^c = \zeta_i \mathbf{v}_i^{c'} \quad (9)$$

and

$$\mathbf{F}_i = \mathbf{A}_i \mathbf{F}_i^c. \quad (10)$$

$\mathbf{A}_i$ is an involutive tensor ($\mathbf{A}_i^{-1} = \mathbf{A}_i$) so that Eq. (1) can be rewritten as

$$\mathbf{v}_i^{c'} + \sum_{j=1}^N \mathbf{T}_{ij}^c \mathbf{F}_j^c = \mathbf{v}_i^c, \quad (11)$$

where

$$\mathbf{T}_{ij}^c = \mathbf{A}_i \mathbf{T}_{ij} \mathbf{A}_j = \begin{pmatrix} T_{ij}^{\theta\theta} & T_{ij}^{\theta r} & T_{ij}^{\theta z} \\ T_{ij}^{r\theta} & T_{ij}^{rr} & T_{ij}^{rz} \\ T_{ij}^{z\theta} & T_{ij}^{zr} & T_{ij}^{zz} \end{pmatrix}. \quad (12)$$

From Eqs. (2) and (11), one can straightforwardly calculate the cylindrical components of the hydrodynamic interaction tensor. The resulting expressions, formulated in Appendix A, are found to be identical with those obtained by García de la Torre and Bloomfield¹⁶ using a rather more indirect method.

For translational motion, the velocity field is homogeneous, so that $\mathbf{v}_i$ (Cartesian) = $\mathbf{v}$ for all beads. By substituting Eq. (9) into (11), one gets

$$\mathbf{F}_i^c + \sum_{j=1}^N \zeta_i \mathbf{T}_{ij}^c \mathbf{F}_j^c = \zeta_i \mathbf{v}_i^c. \quad (13)$$

The shielding tensors in cylindrical coordinates $\mathbf{G}_i^c$ can be defined as usual³:

$$\mathbf{F}_i^c = \zeta_i \mathbf{G}_i^c \mathbf{v}_i^c. \quad (14)$$

From Eqs. (14), (13), and (8) and $\mathbf{A}_i = \mathbf{A}_i^{-1}$, we arrive at the result

$$\mathbf{G}_i^c + \sum_j \zeta_j \mathbf{T}_{ij}^c \mathbf{G}_j^c \mathbf{A}_j \mathbf{A}_i = \mathbf{I}, \quad (15)$$

where $\mathbf{I}$ is the third-rank unit tensor. $\mathbf{G}_i^c$ can be related to $\mathbf{G}_i$ (Cartesian) through Eqs. (14) and (8), the result being

$$\mathbf{G}_i^c = \mathbf{A}_i \mathbf{G}_i \mathbf{A}_i. \quad (16)$$

The translational friction tensor is given by

$$\Xi = \sum_i \zeta_i \mathbf{G}_i = \sum_i \zeta_i \mathbf{A}_i \mathbf{G}_i^c \mathbf{A}_i \quad (17)$$

and the translational friction coefficient

$$f = 3/\text{Tr}(\Xi^{-1}). \quad (18)$$

Symmetric top particles

The qualifier "symmetric top" will be used hereafter for particles having an axis of s -fold symmetry plus a symmetry plane containing this axis. A set of beads which are equivalent from the point of view of rotation about the axis will be denoted as a ring. Beads within a given ring do not necessarily touch each other. Thus, the particle can be regarded as a stack of N_r rings of arbitrary radius. The number of beads in ring k ($k = 1, 2, \dots, N_r$) will be $n_k = s$ except in one case: A single bead whose center lies on the symmetry axis is regarded as a ring with $n_k = 1$.

For symmetric top particles, we have found the following facts:

- (1) $\mathbf{G}_i^c$ is the same for all beads in a given ring.
- (2) All the $\mathbf{G}_i^c$'s have the following form:

$$\mathbf{G}_i^c = \begin{pmatrix} G_i^{\theta\theta} & 0 & 0 \\ 0 & G_i^{rr} & G_i^{rz} \\ 0 & G_i^{rz} & G_i^{zz} \end{pmatrix}. \quad (19)$$

- (3) The Cartesian friction tensor is diagonal, and $\Xi^{yy} = \Xi^{xx}$ when the z axis coincides with the symmetry axis:

$$\Xi = \begin{pmatrix} \Xi^{xx} & 0 & 0 \\ 0 & \Xi^{xx} & 0 \\ 0 & 0 & \Xi^{zz} \end{pmatrix}. \quad (20)$$

- (1) and (3) are evident consequences of symmetry; (3) has been formally demonstrated by Happel and Breener.¹⁷
- (2) is due to the fact that the components of $\mathbf{T}_{ij}^c$ (see Appendix A) are either symmetric or antisymmetric with respect to the transformation $(\theta_j - \theta_i) \rightarrow -(\theta_j - \theta_i)$.

Assume that bead i in Eq. (15) belongs to the k th ring ($i \in k$). Then the summation in Eq. (15) can be decomposed as follows:

$$\sum_{j=1}^{N_r} \zeta_j = \sum_{j \in k} \zeta_j + \sum_{i=1}^{N_r} \sum_{j \in i} \zeta_j. \quad (21)$$

Since $\mathbf{G}_i^c (j \in i) = \mathbf{G}_i^c$, regardless of the value of j within the i th ring, their components will factor out of the last summation on the right-hand side of Eq. (21). After some straightforward manipulation of Eqs. (15), (19),

and (21), we obtain, for $k=1, \dots, N_r$,

$$\sum_{i=1}^{N_r} (M_{ki}^{\theta\theta\theta\theta} G_i^{\theta\theta} + M_{ki}^{\theta\theta rr} G_i^{rr} + M_{ki}^{\theta\theta sr} G_i^{sr}) = 1, \quad (22a)$$

$$\sum_{i=1}^{N_r} (M_{ki}^{rr\theta\theta} G_i^{\theta\theta} + M_{ki}^{rrrr} G_i^{rr} + M_{ki}^{rrsr} G_i^{sr}) = 1, \quad (22b)$$

$$\sum_{i=1}^{N_r} (M_{ki}^{sr\theta\theta} G_i^{\theta\theta} + M_{ki}^{srrr} G_i^{rr} + M_{ki}^{srrs} G_i^{sr}) = 0, \quad (22c)$$

and

$$\sum_{i=1}^{N_r} (M_{ki}^{rrsr} G_i^{sr} + M_{ki}^{rrss} G_i^{ss}) = 0, \quad (23a)$$

$$\sum_{i=1}^{N_r} (M_{ki}^{ssrr} G_i^{rr} + M_{ki}^{ssss} G_i^{ss}) = 1, \quad (23b)$$

where the $M_{ki}^{\alpha\beta\gamma\delta}$'s are formulated in Appendix B. The problem of calculating Ξ is then reduced to solving two sets of linear equations (22) and (23), whose dimensions $3N_r \times 3N_r$ and $2N_r \times 2N_r$ are in general much smaller than $3N \times 3N$. Direct matrix inversion can be used for low N_r , and for higher N_r , the Gauss–Siedel iterative method gives fairly rapid convergence.

From Eqs. (17), (19), and (4), using a decomposition similar to Eq. (21) and applying standard trigonometric formulas, one finally gets

$$\Xi^{xx} = \Xi^{yy} = \sum_{k=1}^{N_r} \zeta_k n_k (G_k^{\theta\theta} + G_k^{rr})/2, \quad (24)$$

$$\Xi^{ss} = \sum_{k=1}^{N_r} \zeta_k n_k G_k^{ss}, \quad (25)$$

and $\Xi^{\alpha\beta} = 0$ for $\alpha \neq \beta$, thus confirming Eq. (20). The translational friction coefficient can be rewritten as

$$f = 3/(2/\Xi^{xx} + 1/\Xi^{ss}). \quad (26)$$

APPLICATION TO CYLINDERS

Calculations

Straight circular cylinders are symmetric top particles, so that Eq. (20) holds for them. Let L and R be the cylinder's length and radius, respectively. It is customary to express the dependence of Ξ^{xx} , Ξ^{ss} , and f on dimensions as

$$\frac{4\eta_0\pi L}{\Xi^{xx}} = \ln p + \gamma_1, \quad (27)$$

$$\frac{2\eta_0\pi L}{\Xi^{ss}} = \ln p + \gamma_{11}, \quad (28)$$

and

$$\frac{3\eta_0\pi L}{f} = \ln p + \gamma, \quad (29)$$

where $p = L/2R$ is the axial ratio. γ_1 and γ_{11} , the end-effect corrections, are functions of p that reach asymptotic values for $p \rightarrow \infty$. From Eq. (26), it follows that $\gamma = (\gamma_1 + \gamma_{11})/2$. There is no exact theory for cylinders. The various treatments proposed so far, including ours, all embody certain degree of approximation, and lead to

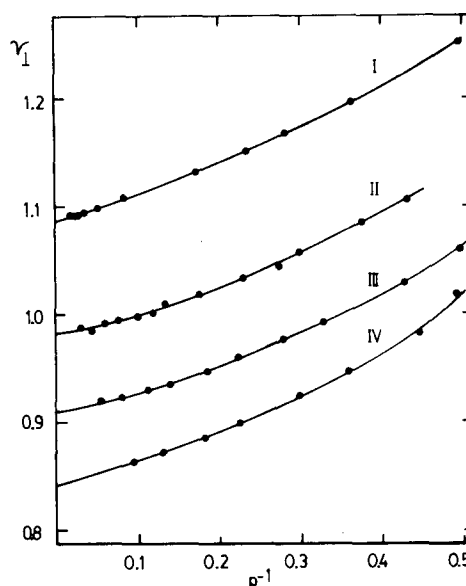


FIG. 1. γ_1 vs p^{-1} for models with (I) $s=3$, mode B; (II) $s=6$, mode A; (III) $s=12$, mode B; (IV) $s=25$, mode A. The continuous curves are the corresponding least-squares fits to Eq. (30).

a variety of values for the end-effects corrections. See the papers by Yamakawa and Tanaka¹⁸ and Nakajima and Wada² for compilations of previous calculations of γ_1 , γ_{11} , and γ .

In this paper, we model a circular cylinder as a stack of N_r rings, each composed of s touching beads of radius σ . Two ways of stacking are considered. In mode A, the centers of the beads of a given ring are vertically above those in the next ring; models with $s=4, 6, 8, 12, 16$, and 25 have been studied. In mode B, each ring is rotated π/s with respect to the next ones in order to get closest packing, and values $s=3, 6$, and 12 were considered. R and L were taken as the circumscribed radius of the model and its total length, respectively. Besides p , another parameter was used to characterize each model: the surface element density λ defined as the ratio of the number of frictional elements sN_r to the area of the cylindrical surface $2\pi RL$. λ only depends on s and the stacking mode. Then, the end-effect corrections can be considered as functions $\gamma_1(p, \lambda)$, $\gamma_{11}(p, \lambda)$, and $\gamma(p, \lambda)$.

In order to interpolate at intermediate values of p and extrapolate to $p \rightarrow \infty$, least-squares second-degree polynomials were used:

$$\gamma_1(p, \lambda) = \gamma_1(\infty, \lambda) + b_1(\lambda)p^{-1} + c_1(\lambda)p^{-2}, \quad (30)$$

$$\gamma_{11}(p, \lambda) = \gamma_{11}(\infty, \lambda) + b_{11}(\lambda)p^{-1} + c_{11}(\lambda)p^{-2}. \quad (31)$$

This functional form has been suggested by analytical calculations such as those reported by Bloomfield *et al.*⁹ and Yamakawa and Fujii,¹⁹ and has been found to be useful in the study of a rigid, double-helical model.²⁰ The fit to Eqs. (30) and (31) is generally good, as shown in Fig. 1 for the γ_1 's in a few cases displayed as examples.

The shell-model limit, corresponding to a perfectly smooth surface, is obtained when $\lambda \rightarrow \infty$. For a sphere of radius R , modeled by small spherical elements of

TABLE I. End-effect corrections in the shell-model limit.

p^{-1}	$\gamma_{\perp}(p, \infty)$	$\gamma_{\parallel}(p, \infty)$	$\gamma(p, \infty)$
0.50	0.99	0.25	0.62
0.46	0.97	0.21	0.59
0.42	0.96	0.18	0.57
0.38	0.94	0.14	0.54
0.33	0.92	0.10	0.51
0.30	0.91	0.07	0.49
0.25	0.90	0.03	0.46
0.20	0.88	-0.02	0.43
0.17	0.87	-0.05	0.41
0.10	0.86	-0.11	0.37
0.06	0.85	-0.15	0.35
0.02	0.84	-0.18	0.33
0.00	0.84	-0.20	0.32

radius βR , Bloomfield *et al.*⁹ found $f/6\pi\eta_0 R = 1 + 0.25\beta$, and this result was later confirmed by McCammon *et al.*²¹ In terms of the surface element density, this equation can be rewritten as $f = 6\pi\eta_0 R + 1.5\eta_0\pi^{-1/2}\lambda^{-1/2}$, which is linear in $\lambda^{-1/2}$. We expect a similar linear relationship to hold for Ξ^{xx} and Ξ^{zz} , and, when p is high enough, for the end-effect corrections too:

$$\gamma_{\perp}(p, \lambda) = \gamma_{\perp}(p, \infty) + d_{\perp}(p)\lambda^{-1/2}, \quad (32)$$

$$\gamma_{\parallel}(p, \lambda) = \gamma_{\parallel}(p, \infty) + d_{\parallel}(p)\lambda^{-1/2}. \quad (33)$$

For the values of p listed in Table I, $\gamma_{\perp}(p, \lambda)$ and $\gamma_{\parallel}(p, \lambda)$ were obtained using the interpolating equations (30) and (31) and, subsequently, extrapolated to $\lambda \rightarrow \infty$ according to Eqs. (32) and (33). These linear extrapolations are quite acceptable, as seen in Fig. 2 for some particular cases. The resulting values of $\gamma_{\parallel}(p, \infty)$, $\gamma_{\perp}(p, \infty)$ and $\gamma(p, \infty)$ are summarized in Table I.

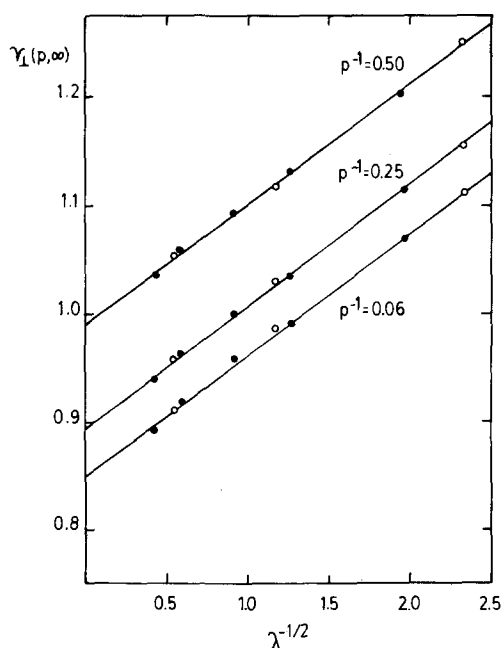


FIG. 2. $\gamma_{\perp}(p, \lambda)$ vs $\lambda^{-1/2}$ for several values of p^{-1} (●), $s=4, 6, 8, 12, 16$, and 25 , mode A; (○) $s=3, 6$, and 12 , mode B. The straight lines are least-squares fits to Eq. (32).

The cylindrical models we have used in the above calculations are hollow and open; they lack solid plates at the ends. To ascertain the influence of the end plates on the translational friction coefficient, we studied two particular cases in which these plates were included. First, models with $s=12$ in mode A were capped with seven beads at each end. The beads were the same size as those used for the cylindrical wall, and arranged in an hexagonal pattern, so that s dropped to 6. In the second case, models with $s=25$ in mode A were capped with solid plates formed by concentric rings of decreasing radius r_k as the center of the plate is approached. The resulting γ 's for $p \approx 1-10$ are listed in Table II along with those for uncapped models. It immediately follows that the contribution of the end plates to the end-effect corrections is rather small for axial ratios as low as 1.0, falling well within the range of numerical uncertainty in calculations or error in experimental data. This situation should also hold in the shell-model limit, and we therefore can conclude that the hollow, uncapped cylinder is a realistic model for solid cylindrical particles when $p > 1$.

Discussion

Our conclusion about the negligible influence of the end plates is in agreement with other studies. Thus, McCammon and Deutch¹ found that the translational friction coefficient does not change appreciably when the hollow core of a cylindrical model ($s=12$, mode A, and $p=0.2-1.03$) was filled with beads. A similar conclusion can be reached by comparison of our results for $s=6$, mode A, with those obtained by Nakajima and Wada² for the corresponding filled model, with $p=0.33-3.33$. This situation can be regarded as a consequence of hydrodynamic interaction. The solvent trapped inside a hollow cylinder must remain immobilized as the cylinder moves, and the portion that is exposed to the external solvent plays a frictional role identical to that of an end plate of a capped cylinder. In fact, we have found that the shielding tensors G_i of beads composing the caps and/or those inside the cylinder are nearly

TABLE II. Values of $\gamma_{\perp}$ and $\gamma_{\parallel}$ for cylindrical models with and without end plates.

n^a	p	With end plates		Without end plates	
		$\gamma_{\perp}$	$\gamma_{\parallel}$	$\gamma_{\perp}$	$\gamma_{\parallel}$
$s=12$, mode A					
5	1.028	1.216	0.604	1.219	0.607
10	2.056	1.049	0.290	1.051	0.291
20	4.112	0.963	0.086	0.964	0.087
43	8.840	0.927	-0.027	0.929	-0.027
$s=25$, mode A					
10	1.114	1.134	0.531	1.143	0.537
20	2.227	0.973	0.217	0.980	0.218
40	4.455	0.893	0.013	0.899	0.015
70	7.796	0.867	-0.071	0.870	-0.075
99	11.026	0.861	-0.110	0.865	-0.109

^aNumber of 12- or 25-sided polygons composing the cylindrical wall.

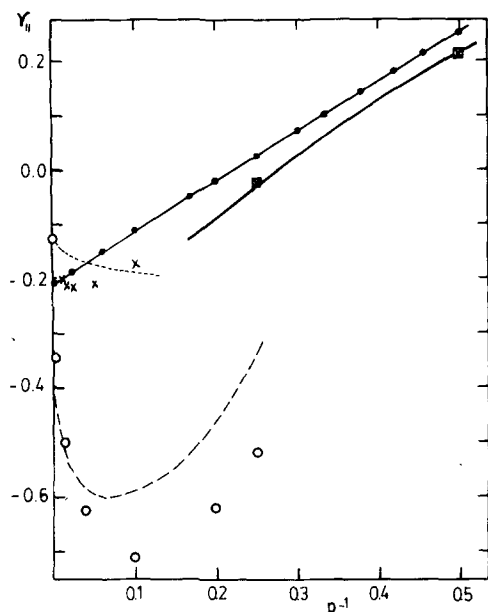


FIG. 3. $\gamma_{\parallel}$ vs p^{-1} . ($\circ$), Broersma, Table III in Ref. 14; (— — —), Broersma, ¹⁴ Eq. (34); (— — — — —), Batchelor, ²⁴ Eq. (35); ($\times$), Younggreen and Acrivos²²; ($\square$), Swanson *et al.*¹⁰; (—), experimental results^{25,17}; (—•—), this work.

zero. This implies that if these beads were absent, the relative velocity at their centers would be also zero.

Once we have demonstrated the adequateness of our models, we can compare their end-effect corrections with those from Broersma's calculation¹⁴ and other theoretical and experimental works. Extrapolation of our results to $p \rightarrow \infty$ yields $\gamma_{\perp}(\infty, \infty) = 0.84$, $\gamma_{\parallel}(\infty, \infty) = -0.20$, and $\gamma(\infty, \infty) = 0.32$, which are slightly different from Broersma's of 0.88, -0.12 , and 0.38, respectively.¹⁴ However, the discrepancy increases dramatically for finite cylinders as p decreases. This is illustrated in Fig. 3, for $\gamma_{\parallel}$, where Broersma's numerical results (Table I in Ref. 14) are represented together with his semiempirical equation

$$\gamma_{\parallel} = \ln 2 - 1.30 + 8[(\ln p)^{-1} - 0.30]^2, \quad 10^5 > p > 4. \quad (34)$$

The sidewise motion of cylinders is axisymmetrical, and several workers have taken advantage of this fact to calculate the corresponding friction coefficient Ξ^{ss} . Thus, Younggreen and Acrivos²² have evaluated numerically Ξ^{ss} for several values of p between 0.5 and 100. Their results turned out to be rather close to others by Gluckman *et al.*²³ Also, Batchelor²⁴ has obtained an analytical expression valid for high p :

$$\gamma_{\parallel} = -0.114 - 0.178(\ln p)^{-1}, \quad p > 10. \quad (35)$$

Moreover, experimental data for both lengthwise and sidewise motion of short macroscopic cylinders with $p < 5$ are available. The experimental curve in Fig. 3 is that reported by Happel and Brenner¹⁷ as a fit of data obtained by Heiss and Coull.²⁵ Finally, there are the calculations of Swanson *et al.*¹⁰ of Ξ^{ss} and Ξ^{ss} for $p = 4$ and 2, that were found to be in excellent agreement with experiments.

As seen in Fig. 3, Broersma's limiting value ($p \rightarrow \infty$)

TABLE III. Errors in Ξ^{ss} and f .^a

p	$\Xi^{ss}/\Xi_{\text{ref}}^{ss}$		f/f_{ref}	
	This work	Broersma	This work	Broersma
∞	1.00	1.00	1.00	1.00
100	1.00	1.07	...	...
10	0.97	1.25	...	...
4	0.95	1.31	0.97	1.30
2	0.96	...	0.95	...

^aBroersma's values¹⁴ at $p \rightarrow \infty$, Younggreen and Acrivos results²² for $p \leq 100$ and experimental results^{25,17} for $p \leq 2$ have been taken for Ξ_{ref}^{ss} .

is confirmed by Batchelor's expression and by the trend of Younggreen and Acrivos' results at high p . However, the discrepancies between Broersma's values for discrete axial ratio and the others results considered here are so remarkable that, in our opinion, the former should be disregarded as representative of the frictional behavior of finite cylinders.

On the other hand, our results for $\gamma_{\parallel}$ are in semiquantitative agreement with other theories and experimental data. The slightly higher values of $\gamma_{\parallel}(p, \infty)$ could be interpreted in terms of the incipient downward curvature that seems to appear at very low $\lambda^{-1/2}$ in the extrapolation to zero bead size (see Fig. 2). Similarly, our $\gamma_{\parallel}(\infty, \infty)$ is smaller than the correct one because we have neglected the upward curvature at very high p . In fact, we detected such a curvature for the models with lowest s , since they can reach highest p , but it did not show up for the higher values of s needed to pass to the shell-model limit.

For $\gamma_{\perp}$, and hence for γ , the situation is analogous to

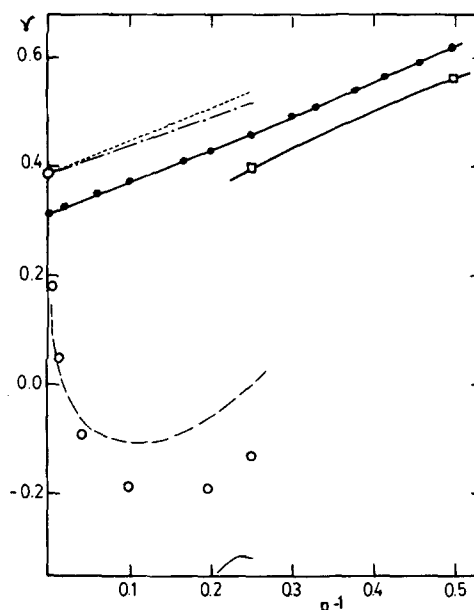


FIG. 4. γ vs p^{-1} . ($\circ$), Broersma, Tables I and III in Ref. 14; (— — —), Broersma, Eqs. (11) and (15) in Ref. 14; ($\circ$), Swanson *et al.*¹⁰; (—), experimental results^{25,17}; (—•—), Yamakawa and Fujii¹⁹; (—•—•—), Bloomfield *et al.*⁹; (—•—), this work.

TABLE IV. Errors in the determination of R from Ξ^{zz} and f , expressed as $R/R_{\text{ref.}}^a$

p	From Ξ^{zz}		From f	
	This work	Broersma	This work	Broersma
∞	0.90	1.00	1.06	1.00
100	1.01	0.75	...	...
10	1.06	0.66	...	...
4	1.07	0.72	1.06	0.67
2	1.04	...	1.06	...

^aWith respect to the reference values specified in Table III.

that described above. In the case of γ , we have two other theoretical approaches, namely, those of Bloomfield *et al.*⁹ and Yamakawa and Fujii,¹⁹ who predicted a linear dependence on p^{-1} at high p , the slopes being $2/\pi$ and $1/2$, respectively. As seen in Fig. 4, the initial slope followed by our results, about 0.57, is an intermediate value close to theirs.

To make a comparative error analysis of our results and Broersma's, we have chosen as reference values Broersma's results at $p \rightarrow \infty$, Younggreen and Acrivos' for $p < 4$, and the experimental curves for $4 \leq p \leq 2$. The errors of the calculated Ξ^{zz} and f are given in Table III as a function of p . It turns out that Broersma's end-effect corrections lead to a significant overestimation of the friction coefficients, while our results deviate 5% at most. In the study of rodlike biopolymers, one often wishes to obtain the hydrodynamic radius of the cylinder from frictional properties, provided that its length has been determined by another technique. In these cases, the use of Broersma's values may cause a strong underestimation of the radius, as shown in Table IV.

CONCLUDING REMARKS

The computational method for symmetric top macromolecules proposed in this paper can be readily applied to a variety of rigid biopolymers that possesses the required symmetry. Thus, we have evaluated the translational friction coefficient of models for T -even bacteriophages⁵ (composed of 421 frictional elements) without having recourse to any simplifying assumption. The results confirmed the validity of the so-called diagonal approximation,³ which was used in a previous study of these particles.⁵ More details of this calculation will be published elsewhere.

Extension of the method to rotational friction coefficient and intrinsic viscosity is not immediate since the symmetry of the flow is not as convenient as in the translational case. However, we have found some symmetry relationships that could be used in a similar manner, and the computational algorithms are presently being developed. The knowledge about rotational friction and intrinsic viscosity of finite right circular cylinders is quite limited, so that a study following the approach suggested in this paper would throw additional light on these aspects of rigid-particle hydrodynamics.

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APPENDIX A

Cylindrical components of the hydrodynamic interaction tensor expressed in the form $(8\pi\eta_0 R_{ij})^{-1} T_{ij}^{\alpha\beta}$, with $\alpha, \beta = \theta, r, z$:

$$\theta\theta: c_1 c_{ij} + c_2 R_{ij}^{-2} r_i r_j s_{ij}^2, \quad (A1)$$

$$\theta r: c_1 s_{ij} + c_2 R_{ij}^{-2} r_j s_{ij} (r_j - r_i c_{ij}), \quad (A2)$$

$$\theta z: c_2 R_{ij}^{-2} z_{ij} r_j s_{ij}, \quad (A3)$$

$$r\theta: -c_1 s_{ij} + c_2 R_{ij}^{-2} r_i s_{ij} (r_j c_{ij} - r_i), \quad (A4)$$

$$rr: c_1 c_{ij} + c_2 R_{ij}^{-2} (r_j - r_i c_{ij})(r_j c_{ij} - r_i), \quad (A5)$$

$$rz: c_2 R_{ij}^{-2} z_{ij} (r_j c_{ij} - r_i), \quad (A6)$$

$$z\theta: c_2 R_{ij}^{-2} z_{ij} r_i s_{ij}, \quad (A7)$$

$$zr: c_2 R_{ij}^{-2} z_{ij} (r_j - r_i c_{ij}), \quad (A8)$$

$$zz: c_1 + c_2 R_{ij}^{-2} z_{ij}^2, \quad (A9)$$

where

$$s_{ij} = \sin(\theta_j - \theta_i), \quad (A10)$$

$$c_{ij} = \cos(\theta_j - \theta_i), \quad (A11)$$

$$z_{ij} = z_j - z_i. \quad (A12)$$

APPENDIX B

The $M_{kl}^{\alpha\beta\gamma\delta}$'s in Eqs. (22) and (23) are given by

$$M_{kl}^{\theta\theta\theta\theta} = \delta_{kl} \left(1 + \zeta_k \sum_{j \in k} c_{ij} T_{ij}^{\theta\theta} \right) + (1 - \delta_{kl}) \zeta_l \sum_{j \in l} c_{ij} T_{ij}^{\theta\theta}, \quad (B1)$$

$$M_{kl}^{\theta\theta rr} = \delta_{kl} \zeta_k \sum_{j \in k} s_{ij} T_{ij}^{\theta r} + (1 - \delta_{kl}) \zeta_l \sum_{j \in l} s_{ij} T_{ij}^{\theta r}, \quad (B2)$$

$$M_{kl}^{\theta\theta zr} = \delta_{kl} \zeta_k \sum_{j \in k} s_{ij} T_{ij}^{\theta z} + (1 - \delta_{kl}) \zeta_l \sum_{j \in l} s_{ij} T_{ij}^{\theta z}, \quad (B3)$$

$$M_{kl}^{rr\theta\theta} = -\delta_{kl} \zeta_k \sum_{j \in k} s_{ij} T_{ij}^{r\theta} - (1 - \delta_{kl}) \zeta_l \sum_{j \in l} s_{ij} T_{ij}^{r\theta}, \quad (B4)$$

$$M_{kl}^{rrrr} = \delta_{kl} \left(1 + \zeta_k \sum_{j \in k} c_{ij} T_{ij}^{rr} \right) + (1 - \delta_{kl}) \zeta_l \sum_{j \in l} c_{ij} T_{ij}^{rr}, \quad (B5)$$

$$M_{kl}^{rr\theta r} = \delta_{kl} \zeta_k \sum_{j \in k} c_{ij} T_{ij}^{r\theta} + (1 - \delta_{kl}) \zeta_l \sum_{j \in l} c_{ij} T_{ij}^{r\theta}, \quad (B6)$$

$$M_{kl}^{rr\theta\theta} = -\delta_{kl}\zeta_k \sum_{j \in k}' s_{ij} T_{ij}^{rr\theta\theta} - (1 - \delta_{kl})\zeta_l \sum_{j \in l}' s_{ij} T_{ij}^{rr\theta\theta}, \quad (B7)$$

$$M_{kl}^{rrrr} = \delta_{kl}\zeta_k \sum_{j \in k}' c_{ij} T_{ij}^{rrrr} + (1 - \delta_{kl})\zeta_l \sum_{j \in l}' c_{ij} T_{ij}^{rrrr}, \quad (B8)$$

$$M_{kl}^{rrrr} = \delta_{kl} \left(1 + \zeta_k \sum_{j \in k}' c_{ij} T_{ij}^{rrrr} \right) + (1 - \delta_{kl})\zeta_l \sum_{j \in l}' c_{ij} T_{ij}^{rrrr}, \quad (B9)$$

$$M_{kl}^{rrrr} = \delta_{kl} \left(1 + \zeta_k \sum_{j \in k}' T_{ij}^{rrrr} \right) + (1 - \delta_{kl})\zeta_l \sum_{j \in l}' T_{ij}^{rrrr}, \quad (B10)$$

$$M_{kl}^{rrrr} = \delta_{kl}\zeta_k \sum_{j \in k}' T_{ij}^{rrrr} + (1 - \delta_{kl})\zeta_l \sum_{j \in l}' T_{ij}^{rrrr}, \quad (B11)$$

$$M_{kl}^{rrrr} = \delta_{kl}\zeta_k \sum_{j \in k}' T_{ij}^{rrrr} + (1 - \delta_{kl})\zeta_l \sum_{j \in l}' T_{ij}^{rrrr}, \quad (B12)$$

$$M_{kl}^{rrrr} = \delta_{kl} \left(1 + \zeta_k \sum_{j \in k}' T_{ij}^{rrrr} \right) + (1 - \delta_{kl})\zeta_l \sum_{j \in l}' T_{ij}^{rrrr}. \quad (B13)$$

The calculated values do not depend on the choice of i within the k th ring. The prime on the summation denotes omission of the term with $i=j$, and $\sum_{j \in k}$ drops off when $n_k = 1$.

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